Flood Risk Modeling in Richmond, BC: A Multi-Scenario Inundation Analysis Using GRASS GIS r.lake

Data Flow

Imput Data

Resolution of 1 meter
UTM Zone 10N (EPSG:3157)

DEM Cleaning

The original DEM included negative or zero values corresponding to water bodies (river, ocean) and no-data cells. These values were removed and replaced with -2 so that r.lake would not interpret the riverbed as solid ground.

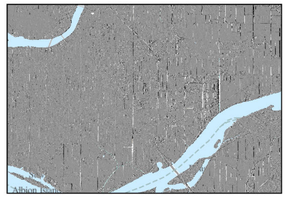
1

Topobathy Creation

DEM that represents both the emerged terrain and an artificial depth for the water body bed.

2

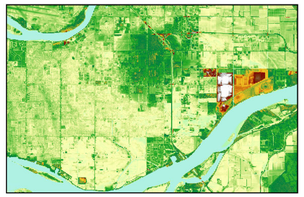
Shaded Relief



A shaded relief map was generated from the DEM using a solar altitude of 45° and an azimuth of 320°.

3

Visual combination

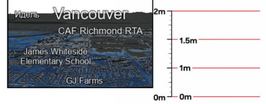


The hillshade was blended with the colorized DEM using a brighten value of 30 to add visual depth to the terrain.

4

Flood modeling

Using r.lake, the extent of a static flood was simulated by filling all DEM cells hydraulically connected to a seed point located on the river. This was performed for four water level scenarios:



A flood raster was generated for each scenario where the cell values represent the water depth.

5

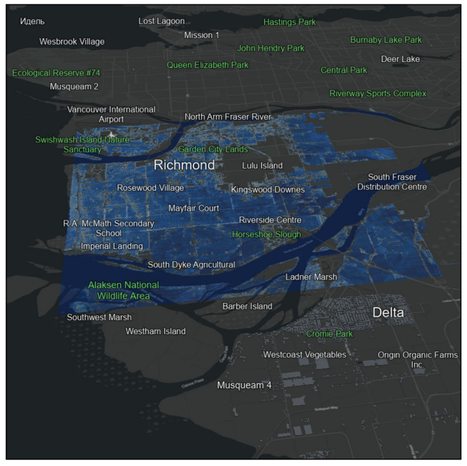
Scenes

No Flooding

1 meter flooding

1,5 meter flooding

2 meter flooding



Analysis

The DEM preprocessing using r.mapcalc.simple was a determining factor for the validity of the model. Without removing zero and negative values, r.lake would have interpreted the Fraser River bed as solid ground, preventing water from propagating correctly from the seed point toward the surrounding urban areas. Assigning -2 meters to null cells ensured a continuous surface with real hydraulic connectivity, an indispensable condition for the filling algorithm to work correctly. An incorrect value at this stage would have produced fragmented or geographically inaccurate flood extents, compromising the reliability of all modeled scenarios.

The application of r.relief.scaling and r.shade does not directly intervene in the flood calculation, but plays a fundamental role in the interpretation of results. The generated hillshade visually highlights the flat topography of the Fraser River delta, a characteristic that explains why small variations in water level produce such significant flood expansion between scenarios. By combining it with the colorized DEM through r.shade, a base map is obtained that clearly identifies the highest-risk zones in relation to their elevation, making the results more communicable and accessible to non-specialized audiences.

The flood simulation was carried out with r.lake, which fills all DEM cells hydraulically connected to a seed point on the Fraser River that fall below a defined water level. Four scenarios were produced: the 0m scenario establishes baseline conditions with no flooding; the 1m scenario captures initial inundation of the lowest areas adjacent to the river; the 1.5m scenario shows significant expansion into urban zones; and the 2m scenario represents the maximum flood extent modeled, exposing the most vulnerable sectors of Richmond. In all output rasters, cell values represent local water depth, providing both the spatial extent and relative severity of each flood event.

3D Scene: https://cogsnscc.maps.arcgis.com/home/item.html?id=f1fd2f61ae5b48fe99f42d95f862118b

Flooding level

High Level

Low Level

N
W E
S

0 3,000 5,000 10,000 20,000 Meters

Projection: NAD 83 UTM Zone 10N
Vertical DATUM: CGVD 2013 v6
Data Source: BC Open Data
Prepared by: Luis Maldonado
Units: Metres
Area of Study: Richmond, British Columbia

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